

Python Programming

Assignment 1

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Objective

The objective of this assignment is to familiarize yourself with the fundamental concepts of Python programming before they are discussed in class.

This is a **reading and exploration assignment**. You are expected to read the official Python documentation, perform small experiments in Jupyter Notebook, and summarize your understanding in **your own words**.

Reference Material

Read the following documentation before attempting the assignment.

- [Python Tutorial – Introduction](#)
- [Python Tutorial – Control Flow](#)

You may also refer to any additional official Python documentation.

Instructions

For each topic listed below, answer the following questions.

1. What is it?
2. Why do we need it?
3. Where and How is it Used?

Write all answers in your own words.

Submission Guidelines

- Prepare the assignment entirely using L^AT_EX.
- Include your Name, Roll Number..
- All code must be executed in Jupyter Notebook and pushed on the github and include a clickable link to your GitHub repository in the report using:

`\href{https://github.com/username/repository}{GitHub Repository}`

- Paste screenshots wherever required.
- Write explanations in your own words.
- Proper formatting and neat presentation will be considered during evaluation.

Topics

Answer what, why and how for all the topics below.

1. Variables
2. Basic Data Types
 - Integer
 - Float
 - Boolean
 - String
3. Arithmetic Operators
4. Comparison Operators
5. Assignment Operators
6. Type Conversion
7. Input and Output
 - `input()`
 - `print()`
8. Strings
 - String Creation
 - Indexing
 - Slicing
 - String Operations
9. Lists
 - Creating Lists
 - Indexing
 - Slicing
 - Updating Elements
 - Nested Lists
 - Basic List Methods
 - `append()`
 - `insert()`
 - `remove()`
 - `pop()`
 - `sort()`
 - `reverse()`
 - `len()`

Mini Experiments

Execute the following programs in Jupyter Notebook.

For each program,

- Write the code.
- Explain the output.

Experiment 1

```
a = 10
b = 20

a, b = b, a

print(a)
print(b)
```

Experiment 2

```
text = "Python Programming"

print(text[0])
print(text[-1])
print(text[0:6])
print(text[::-1])
```

Experiment 3

```
numbers = [10,20,30]

numbers.append(40)
numbers.remove(20)

print(numbers)
```

Experiment 4

```
text = "Programming"

print(text[0:5])
print(text[3:])
print(text[:6])
print(text[::-1])
```

Experiment 5

```
age = input("Enter your age : ")
print(age + age)
age = int(age)
print(age + age)
#Why are the two outputs different?
```

Experiment 6

```
fruits = ["Apple", "Banana", "Mango", "Orange"]
print(fruits[0])
print(fruits[-1])
print(fruits[1:3])
```

Experiment 7

```
numbers = [10,20,30]
numbers[1] = 200
print(numbers)
#Why can a list be modified while a string cannot?
```

Experiment 8

```
print("=" * 40)
print("WELCOME")
print("=" * 40)
print()
print("*" * 20)
print("Python")
```

```
print("*" * 20)
```

Experiment 8

```
# Take an input Name and Reverse reverse it.
```

Experiment 9

```
#Write a Python program that takes the following information from the user:
```

- Name
- Age
- Roll Number
- Course

#Display the information in the following format:

```

""text
=====
                STUDENT CARD
=====
Name           : Alice
Age            : 22
Roll No.       : MS24001
Course         : M.Sc. Scientific Computing
=====

```

Hello Alice!

Welcome to the world of Python.

```
"Python is easy to learn!"
```

'I'll practice every day.'

```
"Practice makes perfect."  
\\//\\//\\//\\//\\//\\//\\//\\//\\//  
Happy Coding!
```

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